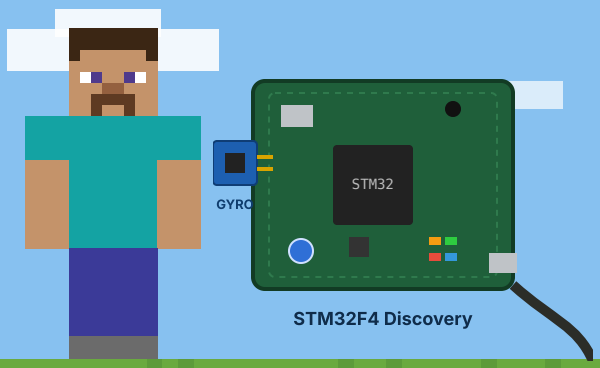
EMBEDDED AI · FINAL PROJECT IDEA

Swing to Mine

A motion controller for Minecraft. Swing the board like a sword or pickaxe, and a neural network running on the chip turns your moves into real key presses.



THE PITCH

Hold an **STM32F4 Discovery board** (plus a $5 gyro) in your hand like a sword. It reads your motion 100 times a second, and a small **int8 neural network on the microcontroller** decides what you just did: slash, mine, place, jump or swap items. The board plugs in over USB and **shows up as a keyboard and mouse**, so Minecraft just works, with no software on the PC.

HOW IT WORKS

01 SENSE
Read motion
Accelerometer + gyro sampled at 100 Hz into a sliding window.

02 THINK
Classify on chip
Pruned int8 CNN runs every ~50 ms on the Cortex-M4. No cloud, no PC.

03 ACT
Press the key
The USB port acts as a keyboard + mouse and sends the input.

04 PLAY
Under 100 ms
Fast enough that the jump lands when you jump.

THE MOVES

Swing sword
One sharp slash
`Left click`

Mine
Keep chopping
`Hold left click`

Place block
Punch forward
`Right click`

Jump
Hop or flick up
`Space`

Swap item
Twist your wrist
`Scroll wheel`

Aim (stretch)
Tilt with the gyro
`Mouse move`

WHY IT'S A GOOD EMBEDDED AI PROJECT

- **Our own dataset:** record every gesture from several people, the same way we did in Lab 2.
- **Fit it on the chip:** prune and int8-quantize with ExecuTorch, as in Lab 3, under a 512 KB flash budget.
- **New challenges:** classify a live stream instead of fixed clips, ignore random movement, and hit a real-time lag budget.
- **Real results:** measure accuracy vs. latency vs. model size, with and without the gyro.

TIMELINE

Nov 6	Idea and team locked in
Nov 12/14	Checkpoint proposal: USB controller working, first gesture data
Late Nov	Train, compress and tune the on-chip model
Dec 8	Live demo: play Minecraft in front of the class

STRETCH GOALS

- Gyro aiming that replaces the mouse
- Second board for the other hand (shield, walking)

Oliver Browne · CU Boulder Embedded AI · Fall 2026

Looking for 1–2 teammates ⛏